

## MATHEMATICAL REVIEW / COMPRESSED PROOF

# A7 curves

## Proof-chain audit and compressed understanding

**Verdict.** The main gonality conclusions survive an independent audit, after correcting the subgroup description and making the numerical positivity step explicit:

$$\lambda_1(C) > 0.34089, \quad \text{gon}(C) \geq 24, \quad \text{gon}(C/\langle\tau\rangle) \geq 12.$$

Here  $C$  is any faithful complex  $A_7$ -curve of signature  $(2, 4, 7)$ , with curvature  $-1$ , and  $\tau$  is any involution. The independent algebraic conclusion  $\text{gon}(C/\tau) \geq 10$  is also sound.

This is a *computer-assisted proof with explicit analytic and arithmetic hypotheses*. It is not an end-to-end Lean proof. The untouched repository needs the repairs described here before its numerical argument is presented as fully justified.

| Proof component          | Audit finding                                                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ramification transport   | Sound. Saturation, centralizer averaging, normalizer generation and cyclic descent give an invariant class of degree 1296, impossible for a free $C_5$ -action. |
| Representation coverage  | Verified from actual Specht characters, rather than character-table orthogonality alone. All irreducibles are covered.                                          |
| Twisted subgroup $Q_2$   | The explicit group is $C_{A_7}((16)(23))$ , of order 24, <i>not</i> $S_4$ . The existing generators and character work.                                         |
| Finite-element bridge    | Sound. A direct energy decomposition proves exactly the lower bound needed, including the constant mode.                                                        |
| Numerical positivity     | All four classes rerun. Independent factor-residual bounds pass for $Q_0, Q_1, Q_2$ , without assuming a sparse Cholesky backward-error theorem.                |
| Refined Castelnuovo step | The two exceptional cases in synthesis Lemma 5.5 are justified in the valid range of Petrakiev's hyperplane-section results.                                    |
| Accessory conclusion     | With the declared classical inputs, every faithful complex $A_7$ -curve has gonality $\geq 19$ , and $\text{ed}_C(A_7; \leq 23) > 1$ .                          |

**The compressed structure.** There are two independent engines. Algebra turns enough forced zeros into an exact Picard identity, then increases its symmetry without adding a factor of five. Analysis turns a finite quadratic-form inequality into a bound for *every* function on the surface. The second engine gives 24/12 directly; it does not depend on the older classification of nine-pencils.

Source snapshot: [JimmysanUWU/ARAGACAS](#).

Branch: `claude/continue-previous-qfhm7j`.

Commit: `8b31a2acf485b25ca4e16067f4862b920452134a`.

Also reviewed: the 19-page *Ramification Transport Synthesis* and the 4-page *Verification Ladder*. No repository files were changed.

1 / THE ALGEBRAIC ENGINE

## Saturation creates a forbidden invariant

Write  $R_v$  for the reduced fixed divisor of an involution. Riemann–Hurwitz and the centralizer formula give

$$g(C) = 136, \quad \deg R_v = 24/2 + 24/4 = 18, \quad g(D) = 64, \quad D = C/\langle \tau \rangle.$$

Orders 3 and 5 act freely. Point stabilizers are cyclic, so distinct commuting involutions cannot fix a common point.

**Determinant lemma.** For a degree- $d$  map  $f$  with  $L = f^*\mathcal{O}_{\mathbb{P}^1}(1)$ , the section

$$s_0 \otimes v^*s_1 - s_1 \otimes v^*s_0 \in H^0(C, L \otimes v^*L)$$

vanishes at every fixed point of  $v$ . It is identically zero exactly when  $f \circ v = f$ . Otherwise its zero divisor has degree  $2d$ . Thus  $\deg R_v > 2d$  forces exact invariance; equality forces  $L + v^*L = [R_v]$ .

On  $D$ , the images of suitable commuting involutions each have 18 fixed points and generate  $C_{A_7}(\tau)/\langle \tau \rangle$ , of order 12. A map of degree  $d \leq 8$  would be invariant under them all, forcing  $12 \mid d$ . Hence  $\text{gon}(D) \geq 9$ .

Suppose  $D$  had a nine-pencil. Its pullback  $f$  to  $C$  has degree 18 and is  $\tau$ -invariant. For  $\mu \neq \tau$  commuting with  $\tau$ ,  $f \circ \mu = f$  is impossible: otherwise  $f$  factors through a Klein group, forcing  $4 \mid 18$ . Its determinant vanishes at  $R_\mu + R_{\tau\mu}$ , exhausting the full degree 36. Therefore

$$L + \mu^*L = [R_\mu + R_{\tau\mu}].$$

Let  $J = C_{A_7}(\tau)$ ,  $|J| = 24$ , and  $M = \sum_{h \in J} h^*L$ . Apply the same identity to every centralizer translate and sum:

$$2M = 24[R_\mu + R_{\tau\mu}], \quad T := 24[R_\tau] + 2M = 24[R_\tau + R_\mu + R_{\tau\mu}].$$

For  $\tau = (12)(34)$ , use the two Klein groups

$$V_0 = \{1, (12)(34), (13)(24), (14)(23)\}, \quad V_1 = \{1, (12)(34), (12)(56), (34)(56)\}.$$

Each normalizer fixes the same class  $T$ . Their normalizers generate  $A_7$ : they contain  $A_4(1234)$ , its conjugate  $A_4(3456)$ , and  $(567)$ . Consequently  $T$  is  $A_7$ -invariant and  $\deg T = 24 \cdot 54 = 1296$ .

An invariant line bundle under a free cyclic action of order  $n$  descends: rescale a generator's bundle isomorphism so its  $n$ th iterate is the identity. Its degree is divisible by  $n$ . Taking  $n = 5$  contradicts  $5 \nmid 1296$ . Thus  $\text{gon}(D) \geq 10$ .

**What makes this work.** The determinant has no residual zeros at saturation. Averaging by 24 removes the pencil choice; normalizers supply the remaining symmetry. There is no cancellation or division of classes in  $\text{Pic}(C)$ , so torsion creates no gap.

## 2 / THE SPECTRAL MODEL

## Every eigenvalue is detected

**Exact surface.** The curve is assembled from 5040 hyperbolic triangles with angles  $\pi/2, \pi/4, \pi/7$ . For a generating pair  $a, b$  of orders 2, 4, tiles  $U_g, L_g$  are joined along  $AB, BC, CA$  to  $L_g, L_{gb}, L_{ga}$ . Vertex cycles have 4, 8, 14 triangles, so there are no cone singularities upstairs. The resulting monodromy is  $(a, b^{-1}, ba)$ ; inversion of  $b$  permutes the original generating pairs. The four inner-conjugacy representatives 0, 1, 12, 14 cover every case.

Every tile uses the same Klein-model triangle and the affine reference domain  $R = \{s, t \geq 0 : s + t \leq 1\}$ . Side identifications preserve the reference coordinates, including for the reflected tiles. Put  $x = X_A + J(s, t)^T$ ,  $J = [X_B - X_A, X_C - X_A]$ . The exact energy and mass coefficients are

$$A_R = |\det J| J^{-1} \frac{I - xx^T}{\sqrt{1 - |x|^2}} J^{-T}, \quad w_R = \frac{|\det J|}{(1 - |x|^2)^{3/2}}.$$

Uniform subdivision therefore gives a genuine mesh of the exact surface; replacing hyperbolic triangles by flat triangles is not part of the proof.

**Representation reduction.** For  $K \leq A_7$  and  $\varepsilon : K \rightarrow \{\pm 1\}$ , restrict to functions satisfying  $f(kx) = \varepsilon(k)f(x)$ . If an eigenvalue's eigenspace contains an irreducible representation whose restriction contains  $\varepsilon$ , that eigenvalue occurs in the twisted problem.

| Problem | Actual subgroup                | Index | Irreducibles detected                |
|---------|--------------------------------|-------|--------------------------------------|
| $Q_0$   | $A_7$                          | 1     | 1, first nonzero eigenvalue          |
| $Q_1$   | $(C_3 \times C_3) \rtimes C_4$ | 70    | 6, $14_a$ , $14_b$ , 15, 21          |
| $Q_2$   | Involution centralizer         | 105   | 10, $\overline{10}$ , 15, 35 (twice) |

The detected dimensions, with multiplicities, sum to 70 and 105 respectively. The nine distinct  $A_7$ -irreducible dimensions satisfy  $\sum_\rho (\dim \rho)^2 = 2520$ . The multiplicities were independently reconstructed from  $S_7$  Specht representations by the Murnaghan–Nakayama rule. The self-conjugate partition  $(4, 1, 1, 1)$  splits into  $10, \overline{10}$  on  $A_7$ ; both have half its character on the subgroup elements, which contain no 7-cycles. This establishes actual representations, not merely an orthogonal numerical table.

**Concrete correction.** Code labels here use letters 0,  $\dots$ , 6. The generators of  $Q_2$  are  $(04)(1362)$  with sign +1, and  $(16)(45)$  with sign  $-1$ . They generate the centralizer of  $(16)(23)$ . It has nontrivial center and elements of order 6;  $S_4$  has neither property. The character is a genuine homomorphism on this actual group. Keep the explicit generators; correct the name.

All irreducibles are detected, so

$$\lambda_1(C) \geq \min\{\mu_2(Q_0), \mu_1(Q_1), \mu_1(Q_2)\}.$$

This step is essential: a large eigenvalue on only one quotient would not bound the full spectrum.

## 3 / THE ANALYTIC BRIDGE

## A finite matrix inequality controls every function

On each reference cell  $e$ , interval bounds give a constant positive matrix  $A_e = c_e P_e \preceq A_R$  and a constant weight  $w_e \geq w_R$ . Let  $a, b$  be the corresponding comparison energy and mass, and  $a_h$  its broken energy on Crouzeix–Raviart (CR) functions. The metric is bounded in Arb at 120-bit precision, over subboxes covering each cell.

Let  $\Pi u$  interpolate the reference-coordinate edge means. If  $e_u = u - \Pi u$ , integration by parts against the constant vector  $A_e \nabla v_h$  gives

$$a_h(e_u, v_h) = 0, \quad a(u) = a_h(\Pi u) + a_h(e_u).$$

This uses zero edge integrals on *every cell*, not a global regularity or conforming-eigenvalue approximation claim. Signs on twisted seams respect the same edge means.

The single-triangle zero-edge-mean inequality [1, Lemma 2.2] implies

$$\|e_u\|_b \leq C_h \sqrt{a_h(e_u)}, \quad C_h^2 \leq \max_e \frac{w_e}{\lambda_{\min}(A_e)} \frac{1}{n^2} \left( \frac{1}{8} + \frac{2}{j_{1,1}^2} \right).$$

Here the chosen edge is the hypotenuse, its opposite vertex has squared maximum distance  $1/n^2$ , and the cell diameter squared is  $2/n^2$ .

Suppose the exact discrete matrices satisfy  $K_h - \sigma M_h \succ 0$ . Then  $\|v_h\|_b \leq \sigma^{-1/2} \sqrt{a_h(v_h)}$ . Triangle inequality and Cauchy–Schwarz now give, for *any* admissible  $u$ ,

$$\begin{aligned} \|u\|_b &\leq \sigma^{-1/2} \sqrt{a_h(\Pi u)} + C_h \sqrt{a_h(e_u)} \\ &\leq \sqrt{\sigma^{-1} + C_h^2} \sqrt{a_h(\Pi u) + a_h(e_u)} = \sqrt{\sigma^{-1} + C_h^2} \sqrt{a(u)}. \end{aligned}$$

Therefore

$$\boxed{\mu_1 \geq \frac{\sigma}{1 + C_h^2 \sigma}}.$$

This direct proof needs neither an approximate eigenvector nor the general higher-eigenvalue theorem.

**The constant mode in  $Q_0$ .** Work with comparison-mean-zero  $u$ , and replace  $\Pi u$  by  $Ju = \Pi u - \text{mean}_b(\Pi u)$ . Its energy is unchanged. The new error is the mean-zero projection of  $e_u$ , so its mass norm is no larger. The same calculation proves the first *nonzero* comparison eigenvalue bound, using the discrete inequality on  $\mathbf{1}^{\perp_M}$ .

For the true metric, energy is larger. For masses, use the weighted variance  $\text{Var}_b(u) = \min_c b(u-c, u-c)$ :  $w_R \leq w_e$  implies  $\text{Var}_{\text{true}}(u) \leq \text{Var}_b(u)$ . Thus the comparison nonzero eigenvalue bound transfers correctly even though the two mean-zero subspaces differ.

**The information content.** Certified discrete energy controls the interpolant; the local estimate controls what interpolation discards. Orthogonality splits the energy exactly. These three facts are the entire finite-to-continuous argument.

## 4 / NUMERICAL RIGOR

## Enclose the constants and factor residuals

**An elementary Bessel bound.** The repository uses a decimal lower bound for the first positive zero  $j_{1,1}$  of  $J_1$ . It is enough to prove  $j_{1,1} > 19/5$ . For  $y = x^2/4 \leq 361/100$ , the alternating Bessel series has the lower bound

$$\frac{J_1(x)}{x/2} \geq P_5(y) := 1 - \frac{y}{2} + \frac{y^2}{12} - \frac{y^3}{144} + \frac{y^4}{2880} - \frac{y^5}{86400}.$$

The omitted alternating tail is positive because its terms decrease. On this interval,

$$P'_5(y) \leq -\frac{1}{2} + \frac{y}{6} - \frac{y^2}{48} + \frac{y^3}{720} \leq -\frac{75235619}{720000000} < 0,$$

using monotonicity of  $y/6 - y^2/48$  for  $0 \leq y \leq 361/100$ . Finally

$$P_5(361/100) = \frac{5335536865199}{864000000000000} > 0.$$

This proves positivity before  $x = 3.8$ , with no root-location oracle.

Taking every interval's upper endpoint before maximizing gives the following safe constants:

| $n$ | Cells per tile | Computed upper bound for $C_h^2$ | Rational bound used |
|-----|----------------|----------------------------------|---------------------|
| 16  | 256            | 0.002190797                      | 1/400               |
| 32  | 1024           | 0.000526892                      | 3/5000              |
| 96  | 9216           | 0.000057107                      | 3/50000             |

**A posteriori positivity lemma.** Let  $B$  be the exact comparison matrix and  $\hat{B}$  its floating approximation, with  $\|B - \hat{B}\|_2 \leq \eta$ . Export a factor  $L$  and its permutation  $P$ . If

$$\|LL^T - P(\hat{B} - cI)P^T\|_2 \leq \delta, \quad c > \delta + \eta,$$

then  $B \succeq (c - \delta - \eta)I \succ 0$ . Indeed  $LL^T \succeq 0$ , and the residual estimate controls everything else. There is *no hypothesis on how  $L$  was computed*.

The independent checker computes the factor product residual in blocks. It uses exact rational arithmetic after converting binary64 endpoints. With  $u = 2^{-53}$ ,  $\gamma_m = mu/(1 - mu)$ ,  $k$  the largest stored row length of  $L$ , and upper bounds  $L_1, L_\infty$  for its absolute norms, it encloses the exact residual by

$$\delta \leq \frac{\rho}{1 - u} + \gamma_{k+1} L_1 L_\infty,$$

where  $\rho$  bounds the computed residual's row and column norms. Positive reductions are enclosed by dividing by  $1 - \gamma_m$ . Diagonal-shift rounding and assembly errors are included separately. The standard binary64 model is assumed; all values are finite, and adding  $10^{-280}$  covers possible underflow terms with ample margin.

For  $Q_0$  use  $B = K - \sigma M + \alpha z z^T$ ,  $z = M\mathbf{1}$ . Positivity implies  $K - \sigma M \succ 0$  on  $\mathbf{1}^{\perp_M}$ . This is precisely the hypothesis needed on the preceding page.

## 5 / INDEPENDENT EVIDENCE

**A conservative rational certificate still gives 24 and 12**

The original full run was reproduced for classes 0, 1, 12, 14. Its reported lower bounds were 0.9978328885 for  $Q_0$ , 0.3408934211 for  $Q_1$ , and 0.6890268661 or 0.7629881031 for  $Q_2$ . These figures were not used as unqualified floating-point eigenvalue estimates.

The independent residual checks passed on all twelve problems. The bottleneck  $Q_1$  has 1,935,360 degrees of freedom and approximately 75 million stored factor entries per class:

| Class | Largest factor row | Computed residual bound $\rho$ | Shift $c$              | Total required bound   |
|-------|--------------------|--------------------------------|------------------------|------------------------|
| 0     | 6501               | $8.95 \cdot 10^{-14}$          | $1.958 \cdot 10^{-10}$ | $9.801 \cdot 10^{-11}$ |
| 1     | 5765               | $1.32 \cdot 10^{-13}$          | $2.064 \cdot 10^{-10}$ | $1.034 \cdot 10^{-10}$ |
| 12    | 6121               | $6.09 \cdot 10^{-14}$          | $2.125 \cdot 10^{-10}$ | $1.064 \cdot 10^{-10}$ |
| 14    | 6346               | $1.29 \cdot 10^{-13}$          | $2.368 \cdot 10^{-10}$ | $1.186 \cdot 10^{-10}$ |

The checker compared exact rational quantities, before rounding these display values. For  $Q_1$ , the true matrix assembly errors were below  $10^{-14}$ . Recovered permutations were checked to be exact bijections. The  $Q_2$  problems have 322,560 degrees of freedom and total required bounds approximately  $3.324\text{--}4.718 \cdot 10^{-11}$ ; every shift is more than 1.99 times its requirement. The  $Q_0$  matrix has 768 degrees of freedom, with shift  $9.891 \cdot 10^{-12}$  and total requirement below  $4.952 \cdot 10^{-12}$ .

**Use rational margins, not final decimal digits.** The matrices establish the stronger discrete thresholds in the original run. They therefore establish these smaller rational thresholds:

| Problem | $n$ | Discrete threshold $\sigma$ | Bound for $C_h^2$ | Continuous lower bound  |
|---------|-----|-----------------------------|-------------------|-------------------------|
| $Q_0$   | 16  | 1                           | 1/400             | 400/401                 |
| $Q_1$   | 96  | 340899/10 <sup>6</sup>      | 3/50000           | 17044950000/50001022697 |
| $Q_2$   | 32  | 17/25                       | 3/5000            | 85000/125051            |

The minimum is the  $Q_1$  fraction, which is  $0.3408920274\dots > 0.34089$ . In particular the result does not rely on interpreting the binary64 value of 0.3409 as an exact decimal.

For a holomorphic degree- $d$  map to  $\mathbb{P}^1$ , Hersch balancing and the Rayleigh principle [2] give

$$\lambda_1(C) \text{Area}(C) \leq 8\pi d.$$

Gauss–Bonnet gives  $\text{Area}(C) = 540\pi$ . Consequently

$$d \geq \frac{135}{2} \lambda_1(C) \geq \frac{1150534125000}{50001022697} > 23.$$

For a degree- $d$  map on  $D = C/\tau$ , compose with  $C \rightarrow D$ :

$$d \geq \frac{135}{4} \lambda_1(C) \geq \frac{575267062500}{50001022697} > 11.$$

Degrees are integers. These inequalities prove  $\text{gon}(C) \geq 24$  and  $\text{gon}(D) \geq 12$ , uniformly over all four classes and all involutions.

## 6 / THE REFINED GEOMETRIC INPUT

## The genus-169 exceptional case is justified

Synthesis Lemma 5.5 excludes independent triples at  $(m, g) = (16, 136)$  and  $(18, 169)$ . Its valid proof uses a hyperplane-section theorem, not the surface theorem that requires ambient dimension at least eight.

Let the eight triple products of three equal degree- $m$  pencils give a birational degree- $d = 3m$  curve  $B \subset \mathbb{P}^7$ . The Equal-Pencil Segre Gap excludes integral surfaces of degree  $\leq 7$  containing  $B$ : since  $d > 14$ , every Segre quadric contains such a surface by Bezout. In  $(\mathbb{P}^1)^3$  its class would have  $a+b+c = 3$ . Type  $(3, 0, 0)$  is reducible,  $(2, 1, 0)$  forces unequal coordinate degrees, and  $(1, 1, 1)$  is a hyperplane, contradicting independence.

**First prove linear normality.** For  $d = 48$ , a ninth section would give a birational model in  $\mathbb{P}^8$  with arithmetic genus at most  $\pi_0(48, 8) = 135 < 136$ . For  $d = 54$ , ten sections give  $\pi_0(54, 9) = 150 < 169$ . Exactly nine sections give a  $\mathbb{P}^8$  curve of genus above  $\pi_1(54, 8) = 156$ ; the Eisenbud–Harris surface theorem [3, Thm. 2.7(a)] supplies a degree-seven surface. Its projection to  $\mathbb{P}^7$  contradicts the Segre gap. A collapsed projection also cannot contain a degree-54 curve. Thus the eight sections are complete and  $B$  is linearly normal.

Its general hyperplane section  $\Gamma \subset \mathbb{P}^6$  is in symmetric position, with  $h_\Gamma(1) = 7$ . Subadditivity and the genus bound give

$$13 \leq h_\Gamma(2) \leq 15.$$

Indeed  $h_\Gamma(2) \geq 16$  forces the sequences

| $d$ | lower Hilbert-function sequence | $p_a(B)$ at most |
|-----|---------------------------------|------------------|
| 48  | 7, 16, 22, 31, 37, 46, 48       | 129              |
| 54  | 7, 16, 22, 31, 37, 46, 52, 54   | 167              |

For  $h_\Gamma(2) = 13$  or  $14$ , the classical Castelnuovo or Eisenbud–Harris lemma gives a curve of degree 6 or 7. For 15, Petrakiev’s Thm. 2.17(a), with parameter  $n = 7$ , applies:  $n \geq 5$  and  $d \geq 2n + 5 = 19$ . Its construction (Lemmas 4.1 and 4.11) gives an integral curve  $D$  of degree at most 8, a component of the quadric base locus.

Degree 8 is impossible. Lemma 2.14 and Remark 2.15 apply with  $\alpha = 2$  and  $q = \lfloor (d-1)/8 \rfloor = 5, 6 \geq 2$ , so the required normality is automatic. They force

$$p_a(B) \leq \pi_2(48, 7) = 127 \quad \text{or} \quad p_a(B) \leq \pi_2(54, 7) = 163,$$

contradicting  $p_a(B) \geq g(C) = 136, 169$ .

Finally linear normality lifts all quadrics through  $\Gamma$  to quadrics through  $B$ . Their base locus has a surface component of degree at most seven, with general hyperplane section  $D$ . It contains  $B$ : otherwise a general hyperplane avoids its finite intersection with  $B$ , while  $\Gamma \subset D$  forces the opposite. This contradicts the Segre gap.

**Conclusion.** Both exceptional exclusions are proved from the precisely identified classical inputs. The dimension restriction on Petrakiev’s Thm. 2.16 is respected; that theorem is not being applied in  $\mathbb{P}^7$ .

## 7 / THE GLOBAL GONALITY CONSEQUENCE

**Every faithful A7-curve has gonality at least 19**

**Global degree-18 exclusion.** For a degree- $m \leq 18$  function on a faithful  $A_7$ -curve, its orbit field is faithful: a trivial action would make  $2520 \mid m$ . The many-functions Castelnuovo bound [4, Lemma 2.2] gives genus at most  $(m-1)^2$ . If that field had index at least two, its functions would have degrees at most nine and its genus at most 64, contradicting the minimum faithful genus 136. Hence the orbit generates the full field and  $g \leq 289$ .

Exact signature enumeration and the natural seven-sheeted quotient leave the following frontier:

| Genus | Signatures           | Bound after this audit |
|-------|----------------------|------------------------|
| 136   | (2, 4, 7)            | gon $\geq 24$          |
| 169   | (3, 3, 5)            | gon $\geq 19$          |
| 199   | (2, 5, 7)            | gon $\geq 19$          |
| 211   | (3, 3, 6), (3, 4, 4) | gon $\geq 19$          |
| 241   | (2, 6, 7), (3, 3, 7) | gon $\geq 19$          |
| 271   | (2, 7, 7)            | gon $\geq 19$          |
| 274   | (3, 4, 5)            | gon $\geq 19$          |

The spectral theorem eliminates the genus-136 row. On every remaining row, two distinct minimum pencils must be birational. Indeed, completing a maximal proper common field of index  $e$  with an orbit function gives

$$g \leq e(m/e - 1)^2 + (e - 1)(m - 1) = m^2/e + em - 3m + 1 \leq 145$$

for  $m \leq 18$  and  $2 \leq e \leq m/2$ , contradicting  $g \geq 169$ .

Three independent pencils give genus at most  $\pi_0(3m, 7)$ . A dependent third is a ratio of two  $(1, 1)$  forms on the first pair's  $(m, m)$  image. Their two base centres have multiplicities  $r, s$  with  $r + s = m$ , costing  $\binom{r}{2} + \binom{s}{2}$  out of the singularity budget  $\delta = (m - 1)^2 - g$ . Thus the general three-pencil bounds for  $m = 14, 15, 16, 17, 18$  are 127, 147, 169, 192, 217. For independent triples at  $m = 15, 17, 18$ , the Segre gap and the first refined bound improve these to 123, 161, 182.

| Genus         | Degrees still to exclude | Final contradiction                                 |
|---------------|--------------------------|-----------------------------------------------------|
| 169           | 14, 15                   | Three-pencil bounds 127, 147                        |
| 169           | 16, 17, 18               | Independent thirds excluded; dependent caps 1, 2, 5 |
| 199, 211      | 16, 17                   | Three-pencil bounds 169, 192                        |
| 199, 211      | 18                       | Independent bound 182; dependent cap 1              |
| 241, 271, 274 | 17, 18                   | Three-pencil bounds 192, 217                        |

Smaller degrees are excluded by  $g \leq (m - 1)^2$ . A minimum pencil has at least eight orbit members: an orbit below seven would be fixed, while an orbit of seven has stabilizer  $A_6$ . Neither simple group acts faithfully on  $\mathbb{P}^1$ , so stabilizing a pencil would force its degree to be divisible by 2520 or 360. Fixing a birational pair therefore leaves at least six thirds.

For the last  $(m, g) = (18, 169)$  case, page 7 excludes independent thirds. The budget is 120. Two disjoint centre pairs cost at least 144, so the pair graph is a star or triangle. Six leaves cost at least 123; a triangle has three edges. At most five thirds fit, contradicting the six in the orbit. The same calculation gives the smaller dependent caps above. This proves the global bound.

## 8 / THE ACCESSORY CONSEQUENCE

**Only degrees 6, 12 and 18 need the geometric obstruction**

**Transfer to accessories.** For a connected accessory of degree  $d \leq 18$ , normalize a faithful linear representation in its compositum with the generic torsor. The synthesis's sweeping transfer lemma, using the icosahedral subgroup  $A_5 \subset A_7$  of order  $60 > d$ , supplies a faithful curve with a function of degree at most  $d$ , impossible by the global bound. The sweep is needed to ensure a component maps nonconstantly. Its norm step selects generic disjoint zero and pole fibres: the norm of an arbitrary chosen function can cancel to a constant.

Here is why the transfer is reliable. Choose a point  $u_0 \in \mathbb{P}^1$  with trivial  $A_5$ -stabilizer and a rational function  $\ell$  equal to one there and zero at the other orbit points. The maps

$$\varphi_v(u) = \sum_{g \in A_5} \ell(g^{-1}u) gv$$

are equivariant and satisfy  $\varphi_v(u_0) = v$ , so they sweep the representation. General pullbacks of the degree- $d$  cover have total component degree  $d$ . A component stabilizer has index at most  $d < 60$  and hence is nontrivial. If every component mapped constantly to the target curve, its image would belong to the finite union of fixed-point sets of nontrivial subgroups. Evaluation at  $u_0$  sweeps the original cover, contradicting dominance. Some component  $Z$  therefore maps nonconstantly to the target and has a degree- $e \leq d$  function.

For its norm, consider the correspondence  $Z \times_T Z$ . Its image in  $\mathbb{P}^1 \times \mathbb{P}^1$  has dimension at most one. Choose zero and pole values outside this correspondence. Their pushforward divisors on  $T$  are disjoint and each has degree  $e$ , so the norm has degree  $e$ . This repairs the unqualified “take the norm of the given function” argument; for example, under  $t = x^2$ , the norm of  $(x-1)/(x+1)$  is constant.

**Prime-local inputs.** The values are  $\text{ed}_{\mathbb{C}}(A_7; 2) = \text{ed}_{\mathbb{C}}(A_7; 3) = 2$  [5]. The Sylow groups are  $D_8$  and  $C_3^2$ , whose minimum faithful complex representation dimensions are two. A prime-to-2 or prime-to-3 extension cannot compress the generic torsor to a curve. Among degrees  $\leq 23$ , only 6, 12, 18 remain.

Let  $F_{\text{acc}}/K$  be such an accessory and  $L/K$  the generic  $A_7$ -extension. The intersection degree  $[F_{\text{acc}} \cap L : K]$  divides  $d$  and is a subgroup index in  $A_7$ . Possible nontrivial indices are 2, 3, 4, 6, 9, 12, 18. Indices at most six are impossible by simplicity and the coset action. For the others:

| index | $ H $ | possible numbers of Sylow 7-groups | $ N_H(C_7) $ |
|-------|-------|------------------------------------|--------------|
| 9     | 280   | 1, 8                               | 280, 35      |
| 12    | 210   | 1, 15                              | 210, 14      |
| 18    | 140   | 1                                  | 140          |

None of the resulting normalizer orders divides  $|N_{A_7}(C_7)| = 21$ . All three accessories are connected; transfer and the global gonality bound now give

$$\boxed{\text{ed}_{\mathbb{C}}(A_7; \leq 23) > 1.}$$

This uses the declared classical inputs and the audited computer-assisted spectral theorem. It makes no claim about unrestricted resolvent towers.

## 9 / REPAIRS AND TRUST BOUNDARY

# What to retain in the final proof

## Mathematical handoff for Claude.

1. Correct  $Q_2$  from “ $S_4$ ” to the explicit involution centralizer in `NOTES.md`, `HANDOFF.md`, `README.md`, and `certify.py`. Keep its generators and character.
2. Replace the unqualified sparse-Cholesky claim by an explicit supported rounding analysis, or preferably the exported-factor residual check described on pages 5–6. Successful Cholesky completion alone is not a certificate.
3. Use a proved Bessel lower bound and conservative rational  $C_h^2$  bounds. The proof on page 5 avoids a hard-coded root approximation.
4. State why the listed characters are genuine and complete. Exact orthogonality alone does not establish this; the Specht construction does.
5. Include the mean-zero projection and weighted-variance argument for  $Q_0$ . The true and comparison mass measures need not have the same mean-zero subspace.
6. Separate the main proof from the superseded nine-pencil classification, approximate spectra, optimality guesses and novelty observations. These are not dependencies of the 24/12 theorem.

**What was checked.** Both manuscripts; the pinned proof and relevant code; group enumeration and normalizer generation; all four triple classes; independent Specht-character multiplicities; exact-geometry and interpolation formulas; the primary triangle estimate; all full certificate outputs; independent sparse and dense factor residuals; and the refined-Castelnuovo ranges and numerical genus bounds. The original `cover.py`, `classreps.py`, `review_checks.py`, and `full_run_certificate.py` passed.

**What is trusted or remains separate.** Classical results explicitly cited below, standard complex-curve geometry and Specht restriction theory, Arb’s enclosure contract, and the standard binary64 dot-product model. The audit inspected the Lean statements and their assumptions, but did not run a Lean build. Lean proves conditional energy/factor implications and arithmetic; it does not consume the interval or matrix certificates. This audit is not a claim of novelty or an independent refereeing of every historical remark.

**Reproduction scope.** Python 3.12.14; NumPy 2.3.5; SciPy 1.17.0; python-flint 0.9.0; CVXOPT 1.3.3; SymPy 1.14.0. Certificate meshes:  $Q_0$  at  $n = 16$ ,  $Q_1$  at  $n = 96$  with binary64 threshold 0.3409,  $Q_2$  at  $n = 32$ . The independent residual checker recovers the CHOLMOD permutation before exporting the factor; `getfactor` consumes the numeric factor. Approximate eigensolvers only choose thresholds on  $Q_2$ ; positivity certifies them.

## Primary references and exact roles.

- [1] C. Carstensen, J. Gedicke, D. Rim, *Explicit error estimates for Courant, Crouzeix–Raviart and Raviart–Thomas finite element methods*, J. Comput. Math. 30 (2012), 337–353. Lemma 2.2 supplies the zero-edge-mean triangle estimate. [Publisher record](#).
- [2] P. C. Yang, S.-T. Yau, *Eigenvalues of the Laplacian of compact Riemann surfaces and minimal submanifolds*, Ann. Sc. Norm. Sup. Pisa 7 (1980), 55–63. Section 2 supplies the degree–area inequality. [Primary text](#).
- [3] I. Petrakiev, *Points in symmetric position and higher Castelnuovo bounds*, arXiv:math/0604517. Theorem 2.7(a), Theorem 2.17(a), Lemma 2.14, Remark 2.15, and the curve/lifting construction give the refined input. [Primary text](#).
- [4] B. Farb, J. Wolfson, *Essential dimension relative to branched covers of degree at most  $n$* , arXiv:2510.22786. Lemma 2.2 is the many-functions genus bound. The sweeping and generic-fibre norm details used here are supplied explicitly by the synthesis. [Author-hosted primary text](#).
- [5] N. Karpenko, A. Merkurjev, *Essential dimension of finite  $p$ -groups*, Invent. Math. 172 (2008), 491–508, Theorem 4.1; with the standard Sylow reduction. [Author-hosted primary text](#).